VASILII ANDREEV

Civil & Structural Engineer · Bridge Engineering · Saint Petersburg, Russia
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RESEARCH PROFILE

Civil and structural engineer with a six-year specialised education in bridge and tunnel engineering and professional experience in structural design, inspection and assessment of existing structures. Research interests: bridge lifecycle management, structural condition assessment, digital twins, predictive maintenance, rehabilitation planning and structural reliability.

EDUCATION

Saint Petersburg State University of Architecture and Civil Engineering (SPbGASU) | 2020–2026
Specialist Degree in Civil & Structural Engineering, with Honours; Bridge and Tunnel Construction. 360 ECTS (ISCED Level 7, Master or equivalent); official GPA 4.93/5.00. Graduate thesis: Major rehabilitation project for the Alarchin Bridge over the Griboyedov Canal; Excellent (A).

Postgraduate studies, SPbGASU | 2026–present
Construction technology and organisation. Research focus: lifecycle management of infrastructure, maintenance decision-making and digital twins.

Peter the Great St. Petersburg Polytechnic University | 2024–2025
Professional retraining — Programming for BIM Platforms (qualification: Programmer).

SPbGASU | 2020–2022
Professional retraining — Translator in the Field of Professional Activity.

RESEARCH EXPERIENCE

Infrastructure lifecycle management & digital twins | 2023–present
Research on assessment algorithms for lifecycle management of construction assets, subsequently specialised for bridge structures; publications covering digital-twin economics, lifecycle management methods and Alarchin Bridge case study. Areas under investigation include the integration of structural data, deterioration prediction and maintenance optimisation. These are research interests, not claims of completed monitoring systems.

SELECTED PUBLICATIONS

1. Andreev, V. K., Kornylev, E. N. “Algorithm for Assessing the Effectiveness and Performance of a Facility Lifecycle Management System for Bridge Structures.” Ekonomika Stroitelstva, issue 7, 2026. [Acceptance confirmed; final bibliographic details to verify].

2. Andreev, V. K., Kornylev, E. N. “Application of the Algorithm for Assessing the Effectiveness and Performance of Bridge Lifecycle Management: The Alarchin Bridge Case Study.” Ekonomika Stroitelstva, issue 8, 2026. [Acceptance confirmed; final bibliographic details to verify].

3. Andreev, V. K. “Digital Twin Technology: Economic Benefits.” Voprosy Studencheskoy Nauki, 1(113), 2026, pp. 481–488.

4. Andreev, V. K. “Digitalisation of Capital Construction in Russia: Lifecycle Management, Standards and Vulnerabilities.” Voprosy Studencheskoy Nauki, 1(113), 2026, pp. 496–502.

5. Andreev, V. K. “Algorithm for Assessing the Effectiveness and Performance of Lifecycle Management of Civil and Industrial Facilities.” Modern Building Materials and Technologies, Kaliningrad, 2026.

6. Andreev, V. K., Nikulin, A. N. “Algorithm for Assessing the Effectiveness and Performance of a Facility Lifecycle Management System.” High Technologies and Innovations, Belgorod, 2025, pp. 343–349.

PROFESSIONAL EXPERIENCE

URBAN LLC — Structural Design Engineer | April 2022–present
Career progression: Technician → Structural Design Engineer, Category III → Category I (July 2026). Structural analysis, calculations, drawings and coordination of steel and reinforced-concrete structures; visual and instrumental surveys, dimensional surveys, defect mapping, verification calculations, condition assessment, and development of repair/strengthening solutions.

SELECTED ENGINEERING WORK

Alarchin Bridge rehabilitation (academic) — Bridge rehabilitation design, FEM and drawings for a historic canal crossing.

Timber chapel (professional) — Structural design and detailed drawing package for timber frames and connections.

Industrial conversion in Petergof (professional) — Steelwork design and analytical checks for industrial adaptation.

Production facility / equipment support (professional) — Structural verification and equipment-support documentation.

Shallow urban tunnel (academic) — Design and engineering calculations for an urban tunnel.

Excavation / PLAXIS (academic) — Staged excavation analysis and assessment of ground/support interaction.

TECHNICAL SKILLS & LANGUAGES

Structural analysis: SOFiSTiK, SCAD Office, LIRA-SAPR, PLAXIS 2D (basic). **CAD/BIM:** AutoCAD, Revit, Tekla Structures, Civil 3D. **Other:** Mathcad, MS Office.
Languages: Russian (native), English (B2 self-assessment), German (A2).

Full engineering project documentation, drawings, and research summaries: see accompanying portfolio website. References available upon request.